

PRODUCTION-READY

Java Full Stack AI Engineer

Complete Syllabus & Engineering Roadmap

A production-focused pathway built from the supplied Full Stack Engineer roadmap, with Java/Spring as the primary backend stack and AI as the final application layer.

Primary Engineering Stack

Java 21 (LTS) | Spring 6 | Spring Boot 3.x | Spring Security 6 | React 18/19 | TypeScript 5+ | PostgreSQL | Kafka | RabbitMQ | Redis | Elasticsearch | Docker | Kubernetes | AWS | OpenTelemetry | Spring AI

Goal: build, secure, test, deploy, observe, scale, and AI-enable enterprise Java full-stack systems.

Source scope: Modules 00-44, engineering workflow, technology stack, and project milestones from the supplied roadmap.

Roadmap Blueprint

How to use this roadmap

Treat every module project as a production artifact. Use Git workflow, code review, tests, security checks, observability, documentation, containerization, and deployment practices throughout the track.

Phase	Focus	Primary Outcome	Source Modules
Phase 0	Engineering Workflow	Company-grade Git, review, branching, SemVer	Workflow
Phase 1	Web & Frontend Foundations	Design, browser, HTML, CSS, JavaScript, TypeScript, React, Next.js	Modules 00-07
Supporting Track	Runtime Interoperability	Node.js, Express.js, MongoDB fundamentals for integration literacy	Modules 08-10
Phase 2	Java + Algorithms	Core Java, concurrency, JVM, DSA	Modules 11-12
Phase 3	Data & Enterprise Java	J2EE, SQL, JDBC, Hibernate/JPA	Modules 13-15
Phase 4	Spring Backend	Spring Core, MVC, Boot, Data JPA, Security/JWT	Modules 16-20
Phase 5	Full-Stack Application Engineering	React + Spring integration, testing, realtime	Modules 21-23
Phase 6	Eventing, Caching & Search	Kafka, RabbitMQ, Redis, Elasticsearch	Modules 24-27
Phase 7	Architecture & Distributed Systems	Patterns, SOLID, Clean, Hexagonal, DDD, System Design, Microservices, Distributed Systems, APIs	Modules 28-36
Phase 8	Production Engineering	Docker, Kubernetes, CI/CD, Cloud, Observability, Performance, AppSec	Modules 37-43
Phase 9	AI Application Engineering	LLMs, Spring AI, RAG, vectors, tools, agents, MCP	Module 44

End-to-End Java Full Stack AI Path

Design -> Web Fundamentals -> React/TypeScript -> Core Java -> DSA -> SQL/JDBC -> Hibernate/JPA -> Spring -> Spring MVC -> Spring Boot -> Security/JWT -> REST/gRPC/GraphQL -> WebSocket -> Kafka/RabbitMQ -> Redis -> Elasticsearch -> Patterns/SOLID -> Clean/Hexagonal/DDD -> System Design -> Microservices -> Testing -> Docker -> Kubernetes -> CI/CD -> AWS -> Observability -> Performance -> AppSec -> Spring AI -> RAG -> Agents/Tools

Technology Coverage

Area	Technologies
Frontend	HTML5, CSS3, JavaScript (ES2024+), TypeScript 5+, React 18/19, Next.js 15+, Redux Toolkit, TanStack Query, React Router v7, Tailwind CSS, React Hook Form, Zod, Jest, RTL, Playwright/Cypress
Java Backend	Core Java 21 (LTS), OOP, Collections, Generics, Records, Sealed Classes, Streams, Lambda, Optional, CompletableFuture, Virtual Threads, Structured Concurrency
Spring	Spring 6, Spring MVC, Spring Boot 3.x, Spring Data JPA, Spring Security 6, JWT, OAuth2, OIDC, Spring Kafka, Spring AMQP, WebSocket/STOMP, SSE
Data	PostgreSQL, MySQL, H2, Hibernate 6, JPA 3.1, Redis, Elasticsearch/OpenSearch, vector databases
Architecture	Design Patterns, SOLID, Clean Architecture, Hexagonal Architecture, DDD, System Design, Microservices, Distributed Systems, API Design
Production	Docker, Kubernetes, Helm, GitHub Actions, Jenkins, ArgoCD/Flux, AWS, Terraform/Pulumi, Prometheus, Grafana, ELK/Loki, Jaeger, OpenTelemetry
AI	OpenAI, Claude, Llama, Gemini, Mistral, Spring AI, LangChain4j/LangChain, RAG, Function Calling, Agents, MCP, vector stores, Ollama/Hugging Face/OpenAI SDK

Engineering Workflow & Git Collaboration (Company-Grade)

Every module project follows this professional workflow.

1. Repository Setup

Step / Type	Action / Naming Convention	Purpose / Example
Fork	GitHub UI -> Fork	Create personal copy
Clone	git clone git@github.com:<you>/fullstack-roadmap.git	Copy to local
Add upstream	git remote add upstream git@github.com:org/fullstack-roadmap.git	Track original
Verify	git remote -v	Confirm remotes
Sync main	git fetch upstream && git checkout main && git merge upstream/main	Stay updated

2. Branch Strategy

Step / Type	Action / Naming Convention	Purpose / Example
main	main	Production-ready, protected
develop	develop	Integration branch
Feature	feature/JIRA-123-user-auth	New features
Bugfix	bugfix/JIRA-456-login-error	Non-critical fixes
Hotfix	hotfix/JIRA-789-prod-down	Critical production
Release	release/v1.2.0	Release preparation
Chore	chore/update-deps	Maintenance
Docs	docs/api-readme	Documentation
Refactor	refactor/service-layer	Code cleanup
Design	design/figma-login-flow	Design assets

3. Commit Conventions

Step / Type	Action / Naming Convention	Purpose / Example
feat	New feature	feat(auth): add JWT refresh token

Step / Type	Action / Naming Convention	Purpose / Example
fix	Bug fix	fix(cart): resolve null pointer
docs	Documentation	docs(readme): update setup steps
style	Formatting	style: apply prettier
refactor	Code refactor	refactor(user): extract mapper
perf	Performance	perf(query): add index hint
test	Tests	test(auth): add login tests
build	Build system	build: bump spring boot to 3.3
ci	CI config	ci: add sonar step
chore	Maintenance	chore: update .gitignore
revert	Revert commit	revert: feat(auth) ...
design	Design assets	design: add login wireframes

4. Code Review & Approval

Step / Type	Action / Naming Convention	Purpose / Example
1	Author	Open PR, assign reviewers
2	CI	Run build, tests, lint, security scan
3	Reviewer	Review code, comment, request changes
4	Designer	Review UI/UX (if applicable)
5	Author	Address feedback, push fixes
6	Reviewer	Approve
7	Maintainer	Merge

5. Review Rules

Rule	Policy
Require PR	main: yes / develop: yes
Required approvals	main: 2 / develop: 1

Rule	Policy
Design approval	main: yes (UI) / develop: yes (UI)
Dismiss stale approvals	main: yes / develop: yes
Require status checks	main: yes / develop: yes
Require conversation resolution	main: yes / develop: yes
Require signed commits	main: yes / develop: no
Require linear history	main: yes / develop: yes

6. Merge Strategies

Step / Type	Action / Naming Convention	Purpose / Example
Squash & Merge	Feature branches (default)	gh pr merge --squash
Rebase & Merge	Linear history preferred	gh pr merge --rebase
Merge Commit	Preserve history	gh pr merge --merge

7. Release & Versioning (SemVer)

Step / Type	Action / Naming Convention	Purpose / Example
MAJOR	Breaking change	1.0.0 -> 2.0.0
MINOR	New feature	1.0.0 -> 1.1.0
PATCH	Bug fix	1.0.0 -> 1.0.1

Foundations

Phase objective

Design and frontend engineering needed to build the user-facing half of enterprise applications.

Module 00 - Design Thinking & Figma

Sub-Moduletopics	
00-01 Design Thinking	Empathize, Define, Ideate, Prototype, Test, Double Diamond, Design Sprint
00-02 UX Research	User Interviews, Surveys, Personas, Empathy Maps, Journey Maps, User Stories
00-03 Information Architecture	Sitemaps, User Flows, Card Sorting, Navigation Patterns
00-04 Wireframing	Low-Fidelity, Mid-Fidelity, High-Fidelity, Sketching, Crazy 8s
00-05 Figma Fundamentals	Frames, Auto Layout, Constraints, Components, Variants, Styles
00-06 Figma Advanced	Design Tokens, Component Sets, Variables, Modes, Prototyping, Smart Animate
00-07 Design Systems	Atomic Design, Tokens, Component Library, Documentation, Storybook
00-08 Prototyping	Clickable Prototypes, Micro-interactions, Transitions, User Testing
00-09 Handoff	Dev Mode, Inspect, Zeplin, Specs, Redlines, Asset Export
00-10 Accessibility Design	Color Contrast (WCAG), Focus States, Touch Targets, Inclusive Design
00-11 Visual Design	Typography, Color Theory, Spacing, Grid Systems, Hierarchy
00-12 Design Critique	Heuristics (Nielsen), Feedback, Iteration, A/B Testing

Project: End-to-End Design of a SaaS Dashboard (Research -> Wireframe -> Hi-Fi -> Prototype -> Handoff)

Module 01 - Internet & Web Fundamentals

Sub-Moduletopics

01-01 How Internet Works

Client-Server, DNS, HTTP/HTTPS, HTTP/2, HTTP/3 (QUIC), TCP/IP, Request-Response, Status Codes, CDN basics

01-02 Web Browsers

Browser Architecture, Rendering Engine, V8, Browser Storage, DevTools, Lighthouse, Core Web Vitals

01-03 Web Security

HTTPS/TLS, CORS, CSP, XSS, CSRF, SQL Injection, Same-Origin, OWASP Top 10

01-04 Web Protocols

REST, GraphQL, gRPC, WebSocket, SSE, WebRTC

01-05 Version Control

Git Internals, Branching, Merge vs Rebase, Cherry-pick, Bisect, Conventional Commits

01-06 Domain & Hosting

Domain Registrar, DNS Records, SSL/TLS Certificates, CDN, Reverse Proxy

Project: Personal Portfolio with Git Workflow (Fork -> PR -> Review -> Merge)

Module 02 - HTML5

Sub-Moduletopics	
02-01 Fundamentals	Document Structure, Semantic Elements, Metadata, SEO, Open Graph
02-02 Forms	Input Types, Validation, ARIA, File Uploads, FormData API
02-03 Multimedia	Audio, Video, Canvas, SVG, iframe, Picture, Source
02-04 Modern APIs	Web Components, Shadow DOM, Templates, Custom Elements, Web Storage, Drag & Drop
02-05 Accessibility	WCAG 2.2, ARIA Roles, Keyboard Nav, Screen Readers, Focus Management
02-06 PWA Basics	Manifest, Service Workers, Offline, Install Prompt

Project: Accessible Multi-Page Website + PWA

Module 03 - CSS3

Sub-Moduletopics	
03-01 Fundamentals	Selectors, Specificity, Box Model, Units, Variables, Cascade, Inheritance
03-02 Layout	Flexbox, Grid, Positioning, Multi-column, Container Queries, Subgrid
03-03 Responsive	Mobile-First, Media Queries, Fluid Typography, Clamp, Container Units
03-04 Modern CSS	Nesting, :has(), :is(), :where(), Layers, Logical Properties, aspect-ratio, accent-color
03-05 Architecture	BEM, Tailwind, CSS Modules, CSS-in-JS, Atomic CSS
03-06 Animation	Transitions, Keyframes, Transform 3D, Scroll-Driven, View Transitions
03-07 Advanced	CSS Houdini, Paint API, Scroll Snap, Custom Properties, color-mix()

Project: Responsive Admin Dashboard with Dark Mode + Animations

Module 04 - JavaScript

Sub-Moduletopics	
04-01 Fundamentals	Variables, Data Types, Coercion, Operators, Control Flow, Strict Mode
04-02 Functions	Declarations, Arrow, Closures, IIFE, HOF, Currying, Memoization
04-03 Objects	Creation Patterns, Prototype Chain, Classes, Getters/Setters, Symbol, WeakMap/WeakSet
04-04 Arrays	map/filter/reduce, Destructuring, Spread/Rest, Iterators, Generators
04-05 Modern JS	Template Literals, Modules, Optional Chaining, Nullish Coalescing, Top-Level Await
04-06 Async	Event Loop, Callbacks, Promises, async/await, Combinators, AbortController
04-07 DOM	Selection, Traversal, Events, Delegation, Forms, MutationObserver
04-08 Browser APIs	Fetch, Storage, IndexedDB, Web Workers, IntersectionObserver, ResizeObserver, Geolocation, Clipboard
04-09 Meta Programming	Proxy, Reflect, Symbol, Decorators
04-10 Error Handling	try/catch, Custom Errors, Global Handlers, Error Boundaries
04-11 Memory	Garbage Collection, Memory Leaks, Heap Snapshots

Project: Task Management SPA (Vanilla JS)

Module 05 - TypeScript

Sub-Moduletopics	
05-01 Fundamentals	Type System, Inference, Interfaces vs Types, Enums, Tuples, any vs unknown
05-02 Advanced Types	Union, Intersection, Generics, Conditional, Mapped, Template Literal, Discriminated Unions
05-03 Utility Types	Partial, Required, Pick, Omit, Record, Readonly, Exclude, Extract, ReturnType
05-04 TypeScript + React	Props, Hooks, Events, Context, Generic Components
05-05 Tooling	tsconfig, Strict Mode, Declaration Files, Type Guards, satisfies, const assertions
05-06 Advanced Patterns	Branded Types, Builder Types, Type-Level Programming

Project: Type-Safe API Client Library

Module 06 - React

Sub-Moduletopics	
06-01 Fundamentals	Architecture, JSX, Components, Props, State, Lifecycle
06-02 Hooks	useState, useEffect, useContext, useReducer, useRef, useMemo, useCallback, useLayoutEffect, useId, useTransition, useDeferredValue, useSyncExternalStore, Custom Hooks
06-03 Patterns	Compound, Render Props, HOC, Controlled/Uncontrolled, Composition, Slots
06-04 State Management	Local, Context, Redux Toolkit, RTK Query, Zustand, TanStack Query, Jotai
06-05 Routing	React Router v7, Nested, Dynamic, Protected, Data Loaders, Lazy
06-06 Forms	Controlled, React Hook Form, Zod, File Uploads, Multi-step
06-07 API	Fetch, Axios, Error Handling, Cancellation, Optimistic Updates, Infinite Queries
06-08 Performance	React.memo, useMemo/useCallback, Code Splitting, Virtualization, Profiler, Concurrent Features
06-09 Testing	Jest, RTL, Hooks Testing, MSW, Integration Tests
06-10 Production	Error Boundaries, Suspense, RSC, Portals, Build Optimization
06-11 Advanced	Server Components, Streaming, Suspense Boundaries, Transitions
06-12 Animation	Framer Motion, React Spring, Layout Animations

Project: Enterprise React Admin Portal (JWT, RBAC, Charts, Realtime)

Module 07 - Next.js

Sub-Moduletopics	
07-01 Fundamentals	App Router, Layouts, Server vs Client Components, File-Based Routing
07-02 Data	Server Components, Server Actions, Route Handlers, Caching, Revalidation, ISR
07-03 Advanced	Middleware, Dynamic, Parallel, Intercepting Routes, Streaming, PPR
07-04 Auth	NextAuth.js, Sessions, Protected Routes, OAuth
07-05 Optimization	Image, Font, Metadata API, SEO, Edge Runtime

Project: Full-Stack Blog with CMS

Java Core & DSA

Phase objective

Master Java language/runtime behavior and the problem-solving patterns needed for production and interviews.

Module 11 - Core Java

Sub-Moduletopics

11-01 Fundamentals	JDK/JRE/JVM, Variables, Types, Operators, Control Flow, Arrays, var
11-02 OOP	Classes, Objects, Encapsulation, Inheritance, Polymorphism, Abstraction, Interfaces
11-03 Advanced OOP	Inner Classes, Static, Final, Packages, Access Modifiers, Records, Sealed Classes, Pattern Matching
11-04 Exceptions	Hierarchy, Checked/Unchecked, Custom, try-with-resources, Multi-catch
11-05 Generics	Classes, Methods, Bounded Types, Wildcards, Type Erasure
11-06 Collections	List, Set, Map, Queue, Deque, Comparable/Comparator, Concurrent Collections
11-07 Functional	Lambda, Functional Interfaces, Method Refs, Streams, Optional
11-08 Concurrency	Threads, Runnable, Callable, Synchronization, Locks, ExecutorService, CompletableFuture, Atomic, Virtual Threads, Structured Concurrency
11-09 I/O & NIO	File I/O, BufferedReader/Writer, Serialization, NIO Channels, Path/Files
11-10 JVM	Memory Model, GC (G1, ZGC), JIT, JVM Tuning, JMX
11-11 Modules (JPMS)	module-info, Requires, Exports
11-12 New Language Features	Text Blocks, Switch Expressions, Records, Sealed Classes

Project: Banking System with Concurrency

Module 12 - Data Structures & Algorithms

Sub-Module topics

12-01 Complexity	Big O, Time/Space, Amortized
12-02 Arrays & Strings	Two Pointers, Sliding Window, Prefix Sum, Kadane, String Manipulation
12-03 Linked Lists	Singly, Doubly, Circular, Fast/Slow, Reversal
12-04 Stacks & Queues	Stack, Queue, Deque, Monotonic Stack, Priority Queue
12-05 Hashing	Hash Tables, Collision, HashMap/HashSet, Frequency
12-06 Trees	Binary, BST, Traversals, AVL, Red-Black, Segment, Fenwick, Trie
12-07 Heaps	Min/Max, Heapify, Heap Sort, Top-K
12-08 Graphs	Representation, BFS, DFS, Topological, Dijkstra, Bellman-Ford, Floyd-Warshall, MST, Union-Find
12-09 Recursion	Patterns, Subsets, Permutations, Combinations, N-Queens, Sudoku
12-10 DP	Memoization, Tabulation, 1D/2D, Knapsack, LCS, LIS, Matrix Chain, DP on Trees
12-11 Advanced	Bit Manipulation, Greedy, Divide & Conquer, Intervals, Segment Trees, Sqrt Decomposition
12-12 Problem Patterns	Sliding Window, Two Pointers, Fast/Slow, Merge Intervals, Cyclic Sort, Top-K, K-way Merge

Practice: 300+ LeetCode Problems

Enterprise Data & Java

Phase objective

Connect Java applications to relational data and understand the classic enterprise Java foundations.

Module 13 - Advanced Java (J2EE)

Sub-Moduletopics	
13-01 JDBC	Architecture, DriverManager, Connection Pooling, PreparedStatement, ResultSet, Batch, Transactions
13-02 Servlets	Lifecycle, Request/Response, Sessions, Cookies, Filters, Listeners, ServletContext
13-03 JSP	Lifecycle, Scriptlets, EL, JSTL, Custom Tags
13-04 MVC	Model, View, Controller, Request Flow

Project: MVC Web Application (Servlet + JSP + JDBC)

Module 14 - SQL & Relational Databases

Sub-Moduletopics	
14-01 Fundamentals	Concepts, Tables, Rows, Columns, Keys, Constraints
14-02 CRUD	SELECT, INSERT, UPDATE, DELETE, UPSERT, MERGE
14-03 Querying	WHERE, ORDER BY, GROUP BY, HAVING, DISTINCT, LIMIT/OFFSET
14-04 Joins	INNER, LEFT/RIGHT, FULL OUTER, CROSS, Self
14-05 Advanced	Subqueries, CTEs, Window Functions, Views, Materialized Views, Stored Procedures, Triggers
14-06 Design	Normalization (1NF-BCNF), ER Diagrams, Relationships, Denormalization
14-07 Transactions	ACID, COMMIT/ROLLBACK, Isolation Levels, Deadlocks
14-08 Performance	Indexes, Query Optimization, EXPLAIN, Partitioning, Connection Pooling
14-09 PostgreSQL	JSON/JSONB, Arrays, Full-Text Search, Extensions, Logical Replication

Project: Enterprise HR Database

Module 15 - Hibernate & JPA

Sub-Moduletopics

15-01 ORM	Concepts, Architecture, Session/SessionFactory, Entity Lifecycle
15-02 Mapping	@Entity, @Table, @Id, @GeneratedValue, @Column, @Transient, @Embeddable
15-03 Relationships	@OneToOne, @OneToMany, @ManyToOne, @ManyToMany, Cascade, Fetch
15-04 Querying	HQL, JPQL, Criteria API, Native, Named Queries
15-05 Advanced	Caching (1st/2nd), Lazy Loading, Fetch Strategies, Batch, Interceptors, Auditing
15-06 Performance	N+1, Fetch Joins, Entity Graphs, Projections, Batching

Project: Employee Management System

Spring Backend

Phase objective

Build maintainable, secure, documented, testable production APIs with the Spring ecosystem.

Module 16 - Spring Framework

Sub-Moduletopics	
16-01 Core	IoC, DI, Beans, Scopes, ApplicationContext, Bean Lifecycle
16-02 Configuration	XML, Java Config, Annotations, Component Scanning, Profiles
16-03 AOP	Aspect, Advice, Pointcuts, Join Points, Proxies
16-04 Data Access	Spring JDBC, Transaction Management, @Transactional, ORM Integration
16-05 Events	ApplicationEvent, @EventListener, Async Events
16-06 Scheduling	@Scheduled, TaskScheduler, Quartz

Project: Spring Core Application

Module 17 - Spring MVC

Sub-Moduletopics	
17-01 Architecture	DispatcherServlet, Handler Mapping, View Resolver, Request Flow
17-02 Controllers	@Controller, @RestController, @RequestMapping, @GetMapping, @PostMapping, @PathVariable, @RequestParam, @RequestBody
17-03 Validation	Bean Validation, @Valid, BindingResult, Custom Validators
17-04 Exception Handling	@ExceptionHandler, @ControllerAdvice, ResponseEntity
17-05 REST	Principles, Methods, Status Codes, Content Negotiation, HATEOAS

Project: Spring MVC REST API

Module 18 - Spring Boot

Sub-Moduletopics	
18-01 Fundamentals	Auto-Configuration, Starters, Embedded Server, CLI
18-02 Config	application.properties/yml, Profiles, @ConfigurationProperties, Env Vars
18-03 REST	Controllers, DTOs, Validation, Exception Handling, HATEOAS
18-04 Production	Actuator, Logging (Logback), Health, Metrics, Graceful Shutdown
18-05 Docs	OpenAPI 3, Swagger, SpringDoc
18-06 Testing	@SpringBootTest, @WebMvcTest, @DataJpaTest, MockMvc
18-07 Advanced	DevTools, Native Image (GraalVM), Virtual Threads, AOT

Project: Production-Ready REST API

Module 19 - Spring Data JPA

Sub-Moduletopics	
19-01 Fundamentals	EntityManager, Persistence Context, Entity States
19-02 Repositories	JpaRepository, CrudRepository, PagingAndSorting, Derived Queries, @Query
19-03 Relationships	All Mappings, Cascade, Orphan Removal
19-04 Advanced	Specifications, Query by Example, Projections, Pagination, Sorting
19-05 Performance	Lazy vs Eager, Fetch Joins, N+1, Entity Graphs, Batch
19-06 Auditing	@CreatedDate, @LastModifiedDate, @CreatedBy
19-07 Soft Delete	@SQLDelete, @Where

Project: Enterprise Backend with JPA

Module 20 - Spring Security & JWT

Sub-Moduletopics	
20-01 Fundamentals	AuthN vs AuthZ, Filter Chain, SecurityContext
20-02 Authentication	UserDetailsService, PasswordEncoder, AuthenticationManager, Provider
20-03 JWT	Structure, Generation, Validation, Filter, Refresh Tokens, Stateless
20-04 Authorization	Roles, Authorities, Method Security, @PreAuthorize, RBAC, ABAC
20-05 OAuth2 & OIDC	Flows, Resource Server, Social Login, Keycloak
20-06 Best Practices	Hashing, CORS, CSRF, Headers, Rate Limiting, mTLS

Project: JWT-Secured Enterprise API

Full Stack & Quality

Phase objective

Integrate React and Spring, then harden the application with realtime features and automated testing.

Module 21 - Full Stack Integration

Layertechnologies

Frontend	React + TS, Redux Toolkit, React Router, RHF, Axios Interceptors
Backend	Spring Boot, Security, JPA, JWT, Validation
Integration	REST, Auth Flow, Error Handling, CORS, Pagination, File Upload, Realtime

Project: Full Stack Enterprise Management Portal (Auth, Employee/Client/Project Mgmt, Kanban, Dashboard, RBAC, Realtime)

Module 22 - Testing

Sub-Moduletopics	
22-01 Unit (Java)	JUnit 5, Assertions, Lifecycle, Parameterized, Test Doubles
22-02 Mocking	Mockito, Mocks/Spies/Stubs, Matchers, Verification
22-03 Spring	@SpringBootTest, @WebMvcTest, @DataJpaTest, MockMvc, TestRestTemplate
22-04 Integration	Testcontainers, PostgreSQL, Kafka, MongoDB
22-05 Frontend	Jest, RTL, User Event, MSW
22-06 E2E	Playwright, Cypress, POM
22-07 API	Postman, REST Assured, Newman
22-08 Contract	Pact, Spring Cloud Contract
22-09 Performance	JMeter, Gatling, k6
22-10 Best Practices	AAA, Isolation, Coverage, Mutation Testing (PIT)
Project: Fully Tested Application (Coverage > 85%)	

Module 23 - WebSocket & Real-Time

Sub-Moduletopics	
23-01 Fundamentals	HTTP vs WS, Lifecycle, Frames
23-02 Spring WS	Config, STOMP, Broker, Topics, @MessageMapping
23-03 Security	Auth, JWT, Authorization
23-04 React	Client, STOMP.js, Realtime State, Reconnection
23-05 SSE	SSE vs WS, Spring SSE, Use Cases
23-06 Scale	Redis Pub/Sub, Broker Relay

Project: Real-Time Chat & Notifications

Messaging & Data Platform

Phase objective

Introduce asynchronous processing, cache design, and full-text/search capabilities.

Module 24 - Apache Kafka

Sub-Moduletopics	
24-01 Fundamentals	EDA, Brokers, Topics, Partitions, Offsets, Consumer Groups
24-02 Producers/Consumers	API, Serialization, Deserialization, Acks
24-03 Spring Kafka	KafkaTemplate, @KafkaListener, Error Handling, Retry, DLT
24-04 Advanced	Partitioning, Ordering, Exactly-Once, Transactions, Kafka Streams
24-05 Ecosystem	Kafka Connect, Schema Registry, ksqlDB
24-06 Patterns	Event Sourcing, CQRS, Saga, Outbox

Project: Event-Driven Order Processing

Module 25 - RabbitMQ & Message Brokers

Sub-Moduletopics	
25-01 Fundamentals	AMQP, Exchanges, Queues, Bindings
25-02 Patterns	Work Queues, Pub/Sub, Routing, Topics, RPC
25-03 Spring AMQP	RabbitTemplate, @RabbitListener
25-04 Reliability	Acks, Nacks, Dead Letter, Idempotency

Project: Task Queue System

Module 26 - Redis & Caching

Sub-Module topics

26-01 Fundamentals

Why, Cache-Aside, Read/Write-Through, Write-Behind

26-02 Redis

Data Structures, Commands, Persistence, Pub/Sub, Streams, Lua

26-03 Spring Cache

@Cacheable, @CacheEvict, @CachePut, CacheManager

26-04 Advanced

TTL, Invalidation, Distributed, Stampede, Redis Cluster

Project: High-Performance Product Catalog

Module 27 - Elasticsearch

Sub-Moduletopics	
27-01 Fundamentals	Index, Document, Mapping, Analyzers
27-02 Querying	Match, Term, Bool, Range, Fuzzy, Aggregations
27-03 Spring Data	Repositories, Template, Criteria
27-04 Advanced	Sharding, Replicas, ILM, Snapshots

Project: Full-Text Search API

Architecture & System Engineering

Phase objective

Move from application-level coding into reusable patterns, boundaries, scale, resilience, and distributed design.

Module 28 - Design Patterns (Complete Catalog)

Sub-Moduletopics	
28-01 Creational	Singleton, Factory Method, Abstract Factory, Builder, Prototype, Object Pool
28-02 Structural	Adapter, Bridge, Composite, Decorator, Facade, Flyweight, Proxy
28-03 Behavioral	Chain of Responsibility, Command, Interpreter, Iterator, Mediator, Memento, Observer, State, Strategy, Template Method, Visitor
28-04 Architectural	MVC, MVP, MVVM, Layered, Microkernel, SOA, Event-Driven, CQRS, Event Sourcing, Saga, Sidecar, Strangler Fig, BFF
28-05 Enterprise	Repository, Unit of Work, Data Mapper, Active Record, Identity Map, Lazy Load, DTO, Value Object, Null Object, Specification
28-06 Concurrency	Producer-Consumer, Reader-Writer, Thread Pool, Future, Actor, Reactor, Proactor
28-07 Cloud	Circuit Breaker, Retry, Bulkhead, Rate Limiter, Timeout, Cache-Aside, Ambassador, Sidecar
28-08 Anti-Patterns	God Object, Spaghetti Code, Big Ball of Mud, Golden Hammer, Premature Optimization

Project: Design Patterns Implementation (Java + Spring + React)

Module 29 - Software Design Principles

Sub-Moduletopics	
29-01 SOLID	SRP, OCP, LSP, ISP, DIP
29-02 Other Principles	DRY, KISS, YAGNI, SoC, Composition over Inheritance, Law of Demeter, Principle of Least Astonishment
29-03 GRASP	Information Expert, Creator, Controller, Low Coupling, High Cohesion, Polymorphism, Pure Fabrication, Indirection, Protected Variations
29-04 Code Quality	Clean Code, Naming, Functions, Comments, Formatting, Refactoring
29-05 Code Smells	Long Method, God Class, Feature Envy, Duplicate Code, Dead Code

Project: Refactoring Legacy Codebase

Module 30 - Clean Architecture

Sub-Module		Topics
30-01 Concepts		SoC, Dependency Rule, Layers, Boundaries
30-02 Structure		domain, application, infrastructure, interface
30-03 Use Cases		Input/Output Ports, Interactors, DTOs
30-04 Benefits		Testability, Maintainability, Framework Independence, Low Coupling

Project: Clean Architecture Spring Boot App

Module 31 - Hexagonal Architecture

Sub-Moduletopics	
31-01 Concepts	Ports & Adapters, Domain Core, App Services
31-02 Ports	Input (Use Cases), Output (Repos, Messaging)
31-03 Adapters	REST, Database, Kafka, External API
31-04 Testing	Mock Adapters, In-Memory Adapters

Project: Hexagonal E-Commerce Backend

Module 32 - Domain-Driven Design

Sub-Moduletopics

32-01 Strategic	Ubiquitous Language, Bounded Contexts, Context Mapping, Core/Supporting/Generic Domains
32-02 Tactical	Entities, Value Objects, Aggregates, Domain Events, Repos, Factories, Domain Services
32-03 Patterns	Aggregate Root, ACL, Repository, Specification, Event Sourcing
32-04 Modeling	Event Storming, Domain Storytelling, Context Mapping Diagrams

Project: DDD-Based Order Management

Module 33 - System Design (Full Module)

Sub-Moduletopics	
33-01 Fundamentals	Scalability, Availability, Reliability, Latency vs Throughput, Vertical vs Horizontal Scaling
33-02 Load Balancing	L4 vs L7, Algorithms (Round Robin, Least Conn, IP Hash, Consistent Hash), HAProxy, Nginx, AWS ALB/NLB
33-03 Caching	Client, CDN, Reverse Proxy, Application, Database; Cache Strategies, Eviction (LRU, LFU, FIFO)
33-04 Databases	SQL vs NoSQL, Replication, Sharding, Partitioning, CAP, PACELC, Indexing, Denormalization
33-05 Consistency	Strong, Eventual, Causal, Read-Your-Writes, Monotonic Reads
33-06 Distributed Systems	Consensus (Raft, Paxos), Leader Election, Distributed Locking, Idempotency, 2PC
33-07 Messaging	Queues, Pub/Sub, Kafka, RabbitMQ, Delivery Guarantees (At-Least-Once, Exactly-Once)
33-08 Storage	Object (S3), Block (EBS), File (EFS), Distributed FS (HDFS)
33-09 CDN & Edge	CDN, Edge Computing, Geo-Distribution, Anycast
33-10 Rate Limiting	Token Bucket, Leaky Bucket, Fixed Window, Sliding Window, Distributed
33-11 API Design	REST, GraphQL, gRPC, Pagination, Versioning, Idempotency
33-12 Observability	Logging, Metrics, Tracing, Alerting, SLI/SLO/SLA
33-13 Case Studies	URL Shortener, Twitter, Instagram, WhatsApp, Netflix, Uber, YouTube, Dropbox, Rate Limiter, Chat, Notification, Autocomplete, Web Crawler, Payment
33-14 Interview Framework	Requirements, Estimation, High-Level Design, Deep Dive, Bottlenecks, Trade-offs

Project: Design 5 Real-World Systems (Document + Diagrams + Trade-offs)

Module 34 - Microservices

Sub-Moduletopics	
34-01 Fundamentals	Monolith vs Modular vs Microservices, Boundaries, Decomposition
34-02 Discovery	Eureka, Consul, Registration
34-03 API Gateway	Spring Cloud Gateway, Routing, Filters, Rate Limiting
34-04 Config	Spring Cloud Config, Centralized, Refresh
34-05 Communication	REST, gRPC, Kafka, RabbitMQ, OpenFeign
34-06 Resilience	Circuit Breaker, Retry, Timeout, Bulkhead, Fallback
34-07 Transactions	Saga, Choreography, Orchestration, Eventual Consistency
34-08 Observability	Tracing (Zipkin/Jaeger), Correlation IDs, ELK
34-09 Service Mesh	Istio, Linkerd, Sidecar, mTLS
34-10 Deployment	Blue-Green, Canary, Rolling

Project: Enterprise Microservices Platform

Module 35 - Distributed Systems

Sub-Moduletopics	
35-01 Fundamentals	Scalability, Availability, Reliability, Fault Tolerance
35-02 CAP	CAP, PACELC
35-03 Consistency	Strong, Eventual, Causal
35-04 Patterns	Idempotency, Distributed Locking, Leader Election, Rate Limiting, Circuit Breaker
35-05 Reliability	Retry + Backoff, DLQ, Health Checks, Graceful Shutdown
35-06 Consensus	Raft, Paxos (concepts)
35-07 Clocks	Lamport, Vector Clocks, Hybrid Logical Clocks

Module 36 - API Design

Sub-Moduletopics	
36-01 REST	Resource URLs, Methods, Status, Pagination, Filtering, Sorting
36-02 Standards	DTOs, Versioning, Error Structure, Idempotency Keys, Rate Limit Headers
36-03 GraphQL	Schema, Queries, Mutations, Resolvers, DataLoader, Subscriptions
36-04 gRPC	Protobuf, Services, Streaming, Interceptors
36-05 Docs	OpenAPI 3.0, Swagger, API Blueprint
36-06 Security	API Keys, OAuth2, JWT, mTLS, HMAC

Project: Well-Designed Public API (REST + GraphQL + gRPC)

Production Engineering

Phase objective

Containerize, automate, deploy, observe, optimize, and secure real systems.

Module 37 - Docker

Sub-Moduletopics	
37-01 Fundamentals	Containers vs VMs, Images, Docker Hub, Dockerfile
37-02 Best Practices	Multi-Stage, Caching, .dockerignore, Security, Distroless
37-03 Compose	Services, Networks, Volumes, Env Vars
37-04 For Java/Node	JVM Containerization, Node, Health Checks, Jib, Buildpacks

Project: Containerized Full Stack App

Module 38 - Kubernetes

Sub-Moduletopics	
38-01 Fundamentals	Architecture, Pods, Services, Deployments, Namespaces
38-02 Config	ConfigMaps, Secrets, Env Vars
38-03 Networking	ClusterIP, NodePort, LoadBalancer, Ingress, Network Policies
38-04 Storage	Volumes, PV, PVC, StorageClass
38-05 Advanced	Helm, HPA, Rolling Updates, Resource Limits, Operators, Service Mesh
38-06 Security	RBAC, Service Accounts, Pod Security, Secrets

Project: Kubernetes-Deployed Microservices

Module 39 - CI/CD & GitOps

Sub-Moduletopics	
39-01 Fundamentals	CI, CD, CD (Deployment)
39-02 GitHub Actions	Workflows, Jobs, Secrets, Matrix, Reusable
39-03 Jenkins	Pipelines, Jenkinsfile, Plugins
39-04 Build Tools	Maven, Gradle, npm/yarn, pnpm
39-05 GitOps	ArgoCD, Flux, Declarative Deployments
39-06 Strategies	Blue-Green, Canary, Rolling, Feature Flags

Project: Complete CI/CD Pipeline with ArgoCD

Module 40 - Cloud (AWS)

Sub-Moduletopics	
40-01 Fundamentals	IaaS/PaaS/SaaS, Regions, AZs
40-02 Compute	EC2, ECS, EKS, Lambda, Fargate
40-03 Storage	S3, EBS, EFS, Glacier
40-04 Database	RDS, Aurora, DynamoDB, ElastiCache
40-05 Networking	VPC, Subnets, SGs, Route 53, ALB/NLB
40-06 DevOps	CodePipeline, CodeBuild, CodeDeploy, CloudFormation
40-07 IaC	Terraform, Pulumi, CDK
40-08 Serverless	Lambda, API Gateway, Step Functions, SAM

Project: Cloud-Deployed Application

Module 41 - Observability

Sub-Moduletopics	
41-01 Logging	Structured, Levels, ELK, Loki
41-02 Metrics	Prometheus, Grafana, Micrometer, Custom
41-03 Tracing	OpenTelemetry, Jaeger, Zipkin
41-04 Alerting	AlertManager, PagerDuty, Slack
41-05 SLO/SLI/SLA	Error Budgets, Golden Signals

Project: Fully Observable Application

Module 42 - Performance Engineering

Sub-Moduletopics	
42-01 Profiling	JProfiler, VisualVM, async-profiler, Chrome DevTools
42-02 JVM Tuning	Heap, GC, JIT, Flags
42-03 Database	Query Plans, Indexes, Partitioning, Connection Pools
42-04 Load Testing	JMeter, Gatling, k6, Locust
42-05 Frontend	Lighthouse, Core Web Vitals, Bundle Analysis

Project: Performance Audit & Optimization Report

Module 43 - Application Security (AppSec)

Sub-Moduletopics	
43-01 OWASP Top 10	Injection, Broken Auth, XSS, CSRF, SSRF, etc.
43-02 Secure Coding	Input Validation, Output Encoding, Parameterized Queries
43-03 Secrets	Vault, AWS Secrets Manager, SOPS
43-04 Scanning	SAST, DAST, SCA, SonarQube, Trivy
43-05 Compliance	GDPR, HIPAA basics
43-06 Cryptography	Symmetric, Asymmetric, Hashing, Salting, Key Management

Project: Security Hardening & Penetration Test Report

AI Integration

Phase objective

Add LLM-powered capabilities to enterprise applications using Spring AI, RAG, tools, and agent patterns.

Module 44 - AI Integration

Sub-Moduletopics	
44-01 LLM Fundamentals	Transformers, Tokenization, Embeddings, Context
44-02 Prompt Engineering	Patterns, Few-Shot, CoT, System Prompts
44-03 Spring AI	ChatClient, EmbeddingClient, Vector Stores, Function Calling
44-04 RAG	Loading, Chunking, Embedding, Search, Context
44-05 Vector DBs	Pinecone, Weaviate, pgvector, Chroma
44-06 Patterns	Function Calling, Agents, Memory, Guardrails
44-07 Advanced	Fine-Tuning, LoRA, MCP, Multi-Agent
44-08 Building	Chatbots, Doc Q&A, Code Assistants, Recommendations

Project: AI-Powered Enterprise Assistant

Supporting Track - Node.js, Express.js & MongoDB

Purpose

These modules remain from the supplied syllabus as interoperability literacy. They are not the primary backend stack of this Java Full Stack AI Engineer pathway.

Module 08 - Node.js

Sub-Module	Topics
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08-01 Fundamentals	Architecture, V8, Event Loop, Non-Blocking I/O, npm, package.json
08-02 Core Modules	fs, path, os, http, https, events, stream, crypto, buffer, worker_threads, child_process
08-03 Async	Callbacks, Promises, async/await, EventEmitter, Streams (Readable/Writable/Transform)
08-04 APIs	File System, HTTP Server, Env Vars, Process, Clustering, Signal Handling
08-05 Performance	Profiling, Diagnostics, Memory Leaks, perf_hooks

Project: CLI Tool & HTTP Server

Module 09 - Express.js

Sub-Moduletopics

09-01 Fundamentals

Setup, Routing, Middleware, Error Handling, Template Engines

09-02 REST

Methods, Status Codes, Content Negotiation, API Versioning

09-03 Architecture

MVC, Routes -> Controllers -> Services -> Repositories, DTOs, Validation

09-04 Security

JWT, bcrypt, OAuth2, CORS, Helmet, Rate Limiting, Sanitization

09-05 Advanced

Multer, Nodemailer, Socket.io, Cron, Winston/Pino, Swagger

Project: Production REST API with Auth

Module 10 - MongoDB

Sub-Moduletopics	
10-01 Fundamentals	NoSQL, Documents, Collections, BSON, Shell, CRUD
10-02 Querying	Operators, Projection, Sort, Pagination, Aggregation, Map-Reduce
10-03 Modeling	Embedded vs Referenced, Patterns, Relationships, Denormalization
10-04 Indexing	Types, Compound, Text, Geospatial, TTL, Explain
10-05 Mongoose	Schemas, Models, Validation, Middleware, Population, Virtuals
10-06 Advanced	ACID Transactions, Replica Sets, Sharding, Change Streams, Time Series
Project: E-Commerce Backend (Node + MongoDB)	

Production-Readiness Gates

Gate	Definition Of Done
Source control	Protected main/develop, PR review, conventional commits, signed commits where required, linear history.
Code quality	SOLID, clean code, refactoring discipline, design patterns where appropriate, documented API contracts.
Security	JWT/OAuth2/OIDC, input validation, secure headers, CORS/CSRF controls, secret management, SAST/DAST/SCA, mTLS where needed.
Testing	Unit, mocking, Spring integration tests, Testcontainers, API tests, contract tests, E2E, performance tests, meaningful coverage.
Reliability	Retries, timeouts, circuit breakers, bulkheads, idempotency, DLQ, graceful shutdown, health checks.
Observability	Structured logs, metrics, traces, correlation IDs, dashboards, alerting, SLI/SLO/SLA and error budgets.
Performance	JVM profiling/tuning, database indexes/plans, connection pools, caching, load testing, frontend bundle/Core Web Vitals.
Delivery	Docker, Kubernetes, CI/CD, GitOps, deployment strategies, infrastructure as code, cloud readiness.
AI safety & operations	Grounding/RAG, tool boundaries, guardrails, model/provider abstraction, evaluation, latency/cost awareness, auditability.

Capstone - AI-Powered Enterprise Platform

Target system

Build an enterprise management platform with a React/TypeScript frontend and Java/Spring Boot backend. Use PostgreSQL for transactional data, Redis for caching, Kafka/RabbitMQ for asynchronous workflows, Elasticsearch for search, WebSocket/SSE for realtime features, Docker/Kubernetes for deployment, AWS for cloud infrastructure, OpenTelemetry for observability, and Spring AI + RAG/vector storage for an enterprise assistant.

Layer	Capstone Scope
Frontend	React + TypeScript, Redux Toolkit/TanStack Query, Router, forms, protected routes, dashboards, realtime UI.

Backend	Spring Boot, REST, validation, centralized errors, Spring Data JPA, transactions, JWT/RBAC, OpenAPI.
Platform	Kafka and/or RabbitMQ, Redis, Elasticsearch, scheduled jobs, file/object storage, background processing.
Architecture	Clean/Hexagonal boundaries, DDD concepts, microservices where justified, gateway/discovery/config, resilience.
Delivery	Docker images, Kubernetes manifests/Helm, CI/CD, GitOps, blue-green/canary/rolling deployment patterns.
Operations	Prometheus/Grafana, logs, distributed tracing, alerting, SLI/SLO/SLA, performance test report.
AI	Spring AI ChatClient, RAG ingestion/chunking/embedding/search, vector store, function calling, agents, guardrails.
Evidence	Architecture document, API specification, test strategy, threat model, dashboards, runbook, deployment guide, trade-off records.

Project Milestones

Milestone	Focus	Expected Artifact
M1	Frontend foundation	Accessible SaaS dashboard + Git workflow
M2	Java foundation	Banking system with concurrency + DSA practice
M3	Persistence	Enterprise HR database + employee management system
M4	Spring backend	Production REST API + JWT-secured enterprise API
M5	Full stack	Enterprise management portal with RBAC, dashboards, realtime
M6	Distributed backend	Event-driven order processing + task queue + cache + search
M7	Architecture	DDD/Hexagonal service + system design case studies + microservices platform
M8	Production	Containerized/Kubernetes deployment + CI/CD + observability + security
M9	AI	AI-powered enterprise assistant with RAG, tools, and evaluation-ready interfaces

Source note: This document is a focused reorganization of the supplied Full Stack Engineer syllabus and roadmap. Module names, topic terminology, projects, and technology coverage are retained from that source unless explicitly framed as a roadmap organization or capstone composition.